Maps between CW Complexes

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CW Complexes

1. Start with a discrete set X^0, whose points are regarded as 0-cells.
2. Inductively, form the n-skeleton X^n from X^{n-1} by attaching n-cells e^n_α via maps $\varphi_\alpha : S^{n-1} \to X^{n-1}$, denoted by $X^n \cup_f X^{n-1}$ (Categorical Pushout)
3. One can either stop this inductive process at a finite stage, setting $X = X^n$ for some $n < \infty$, or one can continue indefinitely, setting $X = \bigcup_n X^n$ (considering the weak topology).

"CW" stands for the following two properties shared by any CW-complex:

> **C = "closure finiteness"**: a compact subset of a CW-complex intersects the interior of only finitely many cells, hence in particular so does the closure of any cell.
>
> **W = "weak topology"**: A subset of a CW-complex is open or closed precisely if its intersection with the closure of each cell is open or closed, respectively.

Whitehead called the interior of the n-disks the "cells", so that their closure of each cell is the corresponding n-disk.

Why do we care about CW complexes?

Good properties

Cell complexes have a very nice mixture of rigidity and flexibility, with enough rigidity to allow many arguments to proceed in a combinatorial cell-by-cell fashion and enough flexibility to allow many natural constructions to be performed on them.

- Locally contractible, normal paracompact and compactly generated.
- Nice categorical properties: quotients, products,
- Cellular approximation theorem, that we will see later.
- **Whitehead's theorem: weak equivalence are homotopy equivalences!**
- Brown's representability theorem: gives necessary and sufficient conditions for a contravariant functor on the homotopy category of pointed connected CW complexes, to the category of sets, to be representable.

Category of CW-complexes

A map $f : X \to Y$ between CW complexes is cellular if $f(X^n) \subset Y^n$ for all $n \geq 0$.

A homotopy $F : X \times I \to Y$ is cellular if it is a cellular map with respect to the natural CW complex structure on $X \times I$.

There are similar definitions for pairs.

Proposition

Let $m > n$. Let U be an open subset of $\mathbb{R}^n$ such that the space $\mathrm{cl}U$ is compact. Let $f : \mathrm{cl}U \to B^m$ be such that $f(\mathrm{fr}U) \subset S^{m-1}$. Then f is homotopic, rel fr U, to a map g such that some point of $B^m - S^{m-1}$ is not in the image of g.

Propostion

Let e^n_α be an n-cell of the CW complex X, and let $z \in \overset{\circ}{e}{}^n_\alpha$. Then $X^n - \overset{\circ}{e}{}^n_\alpha$ is a strong deformation retract of $X^n - \{z\}$.

Cellular aproximation theorem

Theorem

Let $f : X \to Y$ be a map between CW complexes and let A be a subcomplex of X such that $f \mid A : A \to Y$ is cellular. Then f is homotopic, rel A, to a cellular map.

Addendum

The homotopy $H : X \times I \to Y$ obtained in the proof has the property that for each cell e of X, $H(e \times I) \subset C(f(e))$.

Corollary

Let X be a CW complex; X is path connected iff X^1 is path connected.

Now, let's talk about categories!